

# Readme File for Brane Cosmology project

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We consider a bulk spacetime of the form

$$\mathbb{R} \times T^{d_1} \times T^{d_2} \quad (1)$$

where  $\mathbb{R}$  is the time coordinate,  $T^{d_1}$ , and  $T^{d_2}$  are tori of dimensions  $d_1$  and  $d_2$ , respectively. We will wrap a  $d_1$ -brane on the  $T^{d_1}$  tori, and allow the  $d_1$ -brane to move on  $T^{d_2}$ .  $T^{d_1}$  represents the observable Universe, while  $T^{d_2}$  represents two extra dimensions inaccessible to a creature on  $T^{d_1}$ .

We consider an example where  $d_1 = 3$  and  $d_2 = 2$ . We are interested in obtaining the equations of motion for the 3-brane moving along the  $T^2$  tori. Additionally, we consider a light ray being sent along one of the dimensions of the 3-brane and along the two extra dimensions, and sought to obtain the equations of motion for the light ray.

The ordinary differential equations (ODEs) for the 3-brane moving along the two extra dimensions, take the following form:

$$\frac{dy^1}{dt} = \frac{V^{\frac{4}{5}} A^{-\frac{1}{2}} \left( \frac{\tau_1^2 + \tau_2^2 - \tau_1}{\tau_2} \right)}{\sqrt{T^2 V^{\frac{8}{5}} A^{-1} + \frac{\tau_1^2 + \tau_2^2}{\tau_2} p_1 + \frac{p_2^2}{\tau_2} - \frac{2\tau_1 p_1 p_2}{\tau_2}}} \quad (2)$$

$$\frac{dy^2}{dt} = \frac{V^{\frac{4}{5}} A^{-\frac{1}{2}} \left( \frac{p_2 - \tau_1 p_1}{\tau_2} \right)}{\sqrt{T^2 V^{\frac{8}{5}} A^{-1} + \frac{\tau_1^2 + \tau_2^2}{\tau_2} p_1 + \frac{p_2^2}{\tau_2} - \frac{2\tau_1 p_1 p_2}{\tau_2}}} \quad (3)$$

$V(t)$  and  $A(t)$  describes the volume and area as a function of time for the 3-brane.  $V(t)$  and  $A(t)$  are given by the following expressions:

$$V(t) = e^{pt+q} \quad (4)$$

$$A(t) = e^{rt+s} \quad (5)$$

where  $p, q, r$  and  $s$  are constants of integration found by imposing initial conditions. We impose the condition that  $p > 0$ . The  $\tau_1(t)$  and  $\tau_2(t)$  functions describe the geometry of the two extra dimensions.  $\tau_1(t)$  and  $\tau_2(t)$  are related through the following relation:

$$\tau = \tau_1(t) + i\tau_2(t) = \frac{\alpha i e^{ht} + \beta}{\gamma i e^{ht} + \delta} \quad (6)$$

where  $\alpha, \beta, \gamma, \delta$ , and  $h$  are constants of integration. The greek-lettered constants are related by  $\alpha\delta - \beta\gamma = 1$ . For simplicity, we consider a solution where  $\alpha = \delta = 1$  and  $\beta = \gamma = 0$ , so the solution reduces down to the following two equations.

$$\tau_1(t) = 0 \quad (7)$$

$$\tau_2(t) = e^{ht} \quad (8)$$

We require the Hamiltonian of the system to be 0, and can be expressed in the following constraint equation:

$$\frac{4}{5}p^2 = \frac{5}{6}r^2 + \frac{1}{2}h^2 \quad (9)$$

allowing to solve for  $h$ . We notice that the constraint equation yields two real solutions for  $h$ , denoted by  $h_1$  and  $h_2$ , if  $p \gg r$ . Since  $\tau_2(t)$  depends on  $h$ , we will have two ODEs for  $h_1$  and two ODEs for  $h_2$ . Lastly,  $T$ ,  $p_1$ , and  $p_2$  are constants.  $T$  is the energy density of the brane and  $p_1$  and  $p_2$  are the momenta of the brane along the two extra dimensions.

The ODEs for the light ray moving along one of the brane dimensions ( $x^1$ ) and the two extra dimensions ( $x^4$  and  $x^5$ ), are given by the following:

$$\frac{dx^1}{dt} = \frac{V^{\frac{4}{5}} A^{\frac{2}{3}} a_1}{\sqrt{A^{\frac{2}{3}} a_1^2 + \frac{\tau_1^2 + \tau_2^2}{A\tau_2} a_4^2 + \frac{a_5^2 - 2\tau_1}{A\tau_2}}} \quad (10)$$

$$\frac{dx^4}{dt} = \frac{V^{\frac{4}{5}} \left( \frac{(\tau_1^2 + \tau_2^2) a_4 - \tau_1 a_5}{A\tau_2} \right)}{\sqrt{A^{\frac{2}{3}} a_1^2 + \frac{\tau_1^2 + \tau_2^2}{A\tau_2} a_4^2 + \frac{a_5^2 - 2\tau_1}{A\tau_2}}} \quad (11)$$

$$\frac{dx^5}{dt} = \frac{V^{\frac{4}{5}} \left( \frac{-\tau_1 a_4 + a_5}{A\tau_2} \right)}{\sqrt{A^{\frac{2}{3}} a_1^2 + \frac{\tau_1^2 + \tau_2^2}{A\tau_2} a_4^2 + \frac{a_5^2 - 2\tau_1}{A\tau_2}}} \quad (12)$$

where  $a_1$  is the momenta of the light ray along one of the brane dimensions and  $a_4$  and  $a_5$  are the momenta of the light ray along the two extra dimensions.  $a_1$ ,  $a_4$ , and  $a_5$  are constants and have the following constraint.

$$\sqrt{a_1^2 + a_4^2 + a_5^2} = 1 \quad (13)$$

Again, since  $\tau_2(t)$  depends on  $h$ , we will have three ODEs for  $h_1$  and three ODEs for  $h_2$ .

The program uses both the Euler's method and fourth-order Runge-Kutta to solve the 4 ODEs for the brane and 6 ODEs for the light ray. The program

plots the solutions using matplotlib. The code begins by prompting the user to input the initial conditions for the problem (lines 13-28, 37-38, and 40-42), and stores them in variables. In addition, we must specify the brane (lines 30-31) and light ray (lines 33-34) initial positions at  $t_{ini}$  for both  $h_1$  and  $h_2$ . We store these as arrays, so change the arrays for desired initial positions.

We now discuss a specific example for a first-time user to run. Before running the program ensure you have numpy, matplotlib, and SymPy<sup>1</sup> on your machine. Run the program and respond to the prompts. Enter the following numbers without no spaces in the following order: 0, 10, 20, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 10, 0.001, and 0.0001. In this specific example, we are defining the initial conditions for  $V(t)$  and  $A(t)$  as follows.

$$V(0) = 10 \tag{14}$$

$$V'(0) = 20 \tag{15}$$

$$A(0) = A'(0) = 1 \tag{16}$$

We give the brane momentum along one of the extra dimensions ( $p_1 = 1$  and  $p_2 = 0$ ). The energy density is given by  $T = 1$ . We give the light ray momentum only along one of the brane dimensions and not the two extra dimensions ( $a_1 = 1$ ,  $a_4 = 0$ , and  $a_5 = 0$ ). For this example, we start at  $t_{ini} = 0$ , and evolve to  $t_{end} = 10$ . For the Euler's method, we choose the time step to be 0.001 and for the fourth-order Runge-Kutta we choose the time step to be 0.0001.

For this specific example, we obtain a plot showing the solutions to the four ODEs for the brane motion, and another plot for the solutions to the six ODEs for the light ray motion. Refer to each figure for a description of the solutions to the ODEs for both scenarios.

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<sup>1</sup><https://www.sympy.org/en/index.html>

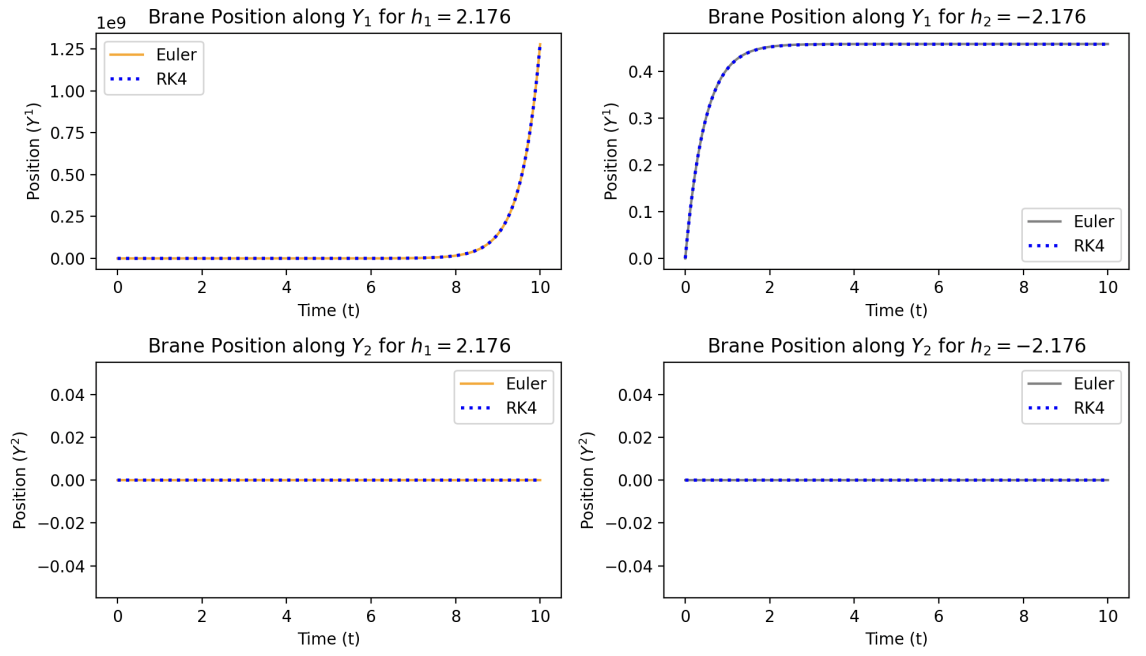


Figure 1: Plot showing the brane position solutions (using Euler's method and fourth-order Runge-Kutta) to the two ODEs for  $h_1$  and the two ODEs for  $h_2$ . We notice that the the brane position along  $Y_2$  remains at 0. This makes sense because we did not prescribe momentum along that direction ( $p_2 = 0$ ).

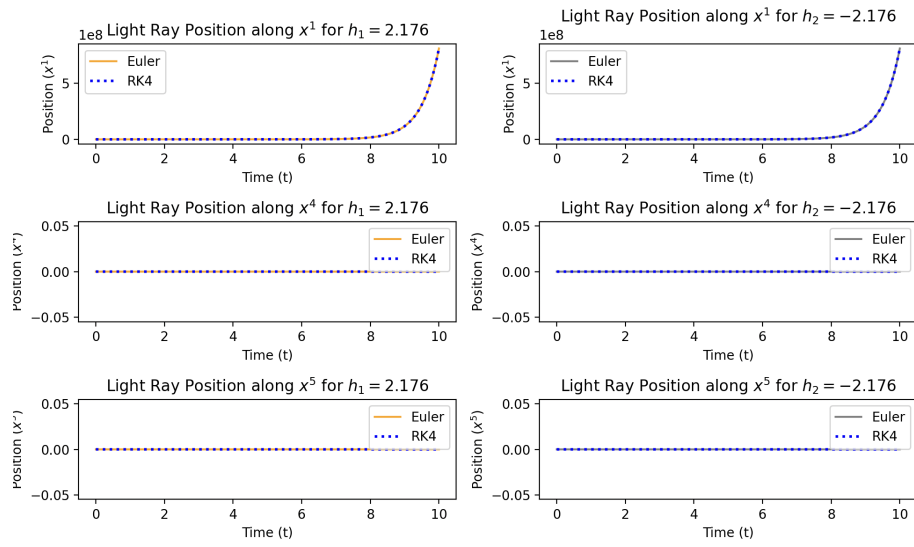


Figure 2: Plot showing the light ray position solutions (using Euler's method and fourth-order Runge-Kutta) to the three ODEs for  $h_1$  and the three ODEs for  $h_2$ . We notice that the the light ray position along  $x_4$  and  $x_5$  remains at 0. This makes sense because we did not prescribe momentum along those two directions ( $a_4 = a_5 = 0$ ).